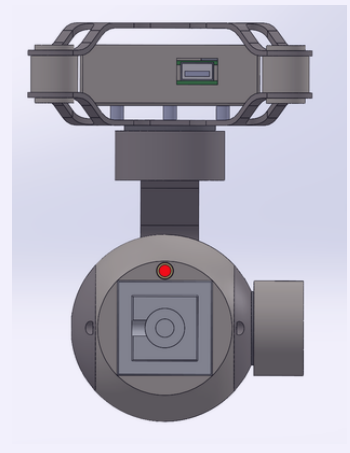
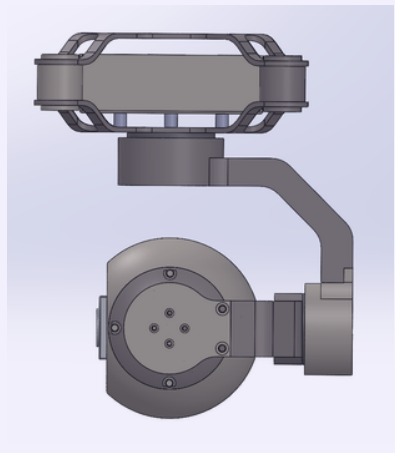
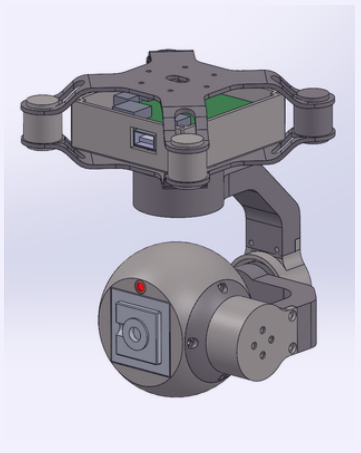
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3-AXIS DRONE CAMERA GIMBAL - UTSA UNMANNED SYSTEMS LAB


What?

- Design and fabricate the shell of a **self stabilizing drone camera gimbal** with **3 axis rotation**
- Create a product that is **designed for efficient manufacturability and assembly**
- House all necessary circuit boards, modules, and wiring

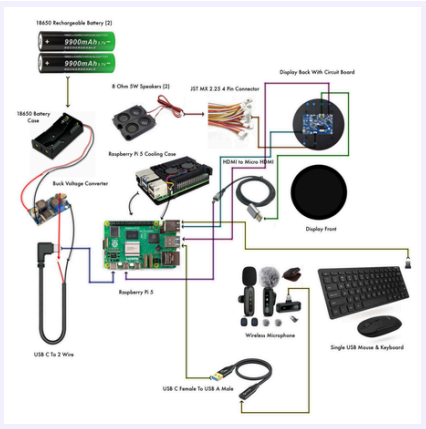
How?

- Used **SolidWorks** to design and **3D Printing** for rapid prototyping
- **Brushless DC motors** used for control of arms
- **Weekly team design meetings** were held to discuss possible changes/improvements

Results

- After multiple prototypes, a **final prototype** has been developed and **implemented in drones** showing real-world functionality
- **Fully self-stabilizing functionalities**
- Full **3-degrees of freedom** movement
- 4k camera and laser pointer

NOVA POD - PERSONAL PROJECT








What?

- Design and program an **AI Home Assistant**
- Emphasis on **human-like conversational interactions** between the user and the assistant
- Develop an **aesthetically pleasing shell** and **interactive GUI** for the display

How?

- Designed the shell in **SolidWorks** to fit around the display and house components
- Designed the **system architecture** to ensure everything works properly
- Used **Python** to program the conversational element, utilizing **OpenAI and ElevenLabs APIs** for realistic interaction
- GUI developed with Python using **Pygame**

Results

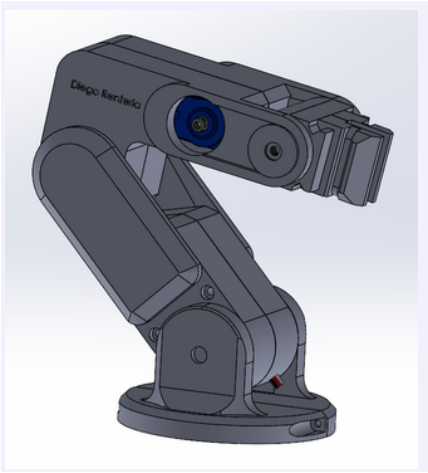
- **Human-sounding conversational AI assistant** that can answer virtually any question and **remember context** from earlier conversations
- Functional and aesthetically pleasing **interactive GUI** with four apps **(Weather, News, To-Do, Alarm)**
- **Sleek spherical shell** design for modern look

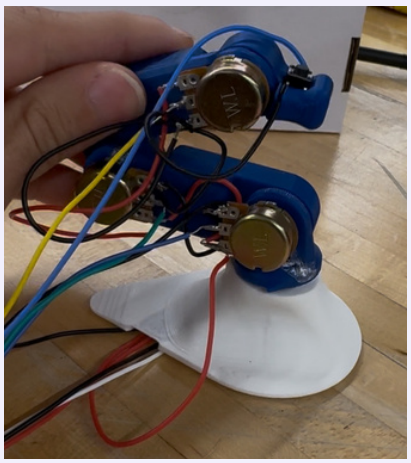
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ROBOT ARM - PERSONAL PROJECT







What?

- Create a **robot arm with a gripper** that can be controlled to **pick up small light objects**
- Make **control of the robot simple and intuitive** for the user
- Use an **Arduino** to code and program the arm

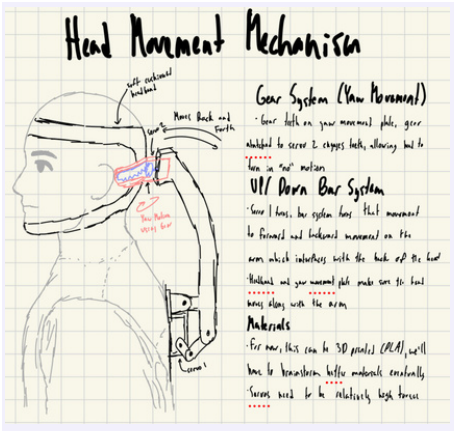
How?

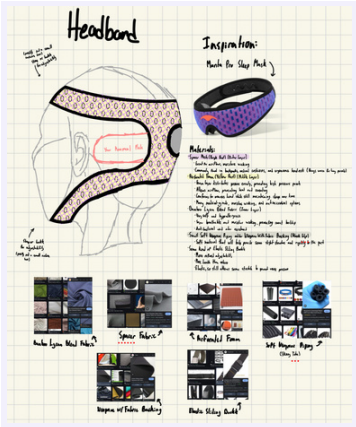
- Used a **mini version with potentiometers** to control the main arm, where the **main arm's servo motors match the mini arm's positions.**
- Designed all of the parts in **SolidWorks**, used **5 servo motors** to control the main arm, and **4 potentiometers** to give the position of the small arm
- **Soldered** all components and wires together
- Using **Arduino IDE** to program the arm

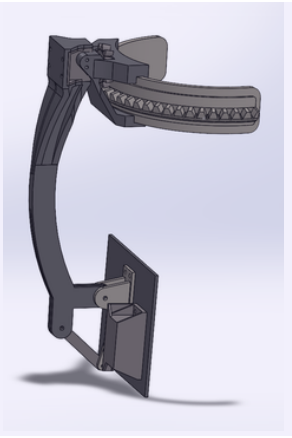
Results

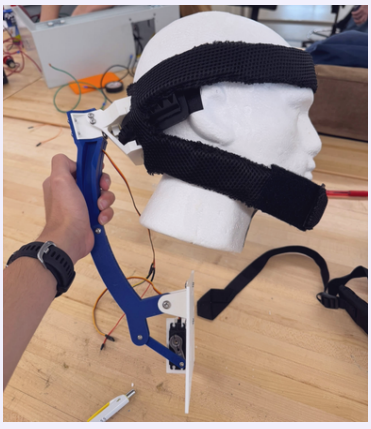
- Functional robot arm that can pick up and reposition small objects
- Large arm copies the "master" arm through potentiometer movement input

EXOSKELETON HEAD MOVEMENT - EPICS CLUB









What?

- Create a **mechanical exoskeleton system** to allow people with ALS to gain a wider range of motion in their head movements
- Emphasis on **comfort over long periods of wear**

How?

- **Brainstormed** possible **approaches and materials**
- **SolidWorks** to design pitch single bar linkage system and **3D Printed** the parts
- Arduino to control the servos
- **Gave every member of the group a role** to ensure we finish on time

Results

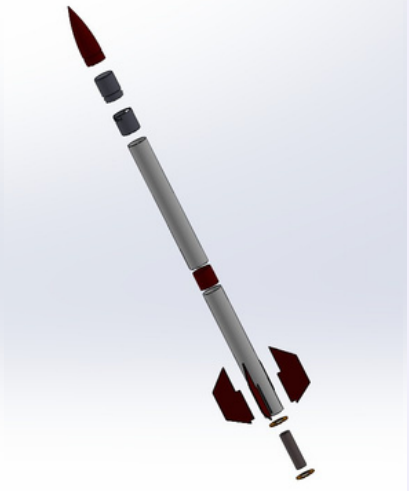
- **Functional neck exoskeleton prototype** with yaw and roll movement of the head
- Motors **controlled using joystick** through an Arduino Uno

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MONSTER MASH ROCKET - AERONAUTICS AND ROCKET CLUB







What?

- Design and fabricate a rocket that safely reaches an **apogee of approximately 1000ft**
- Design a **parachute recovery** system to safely recover the rocket with **minimal damages**

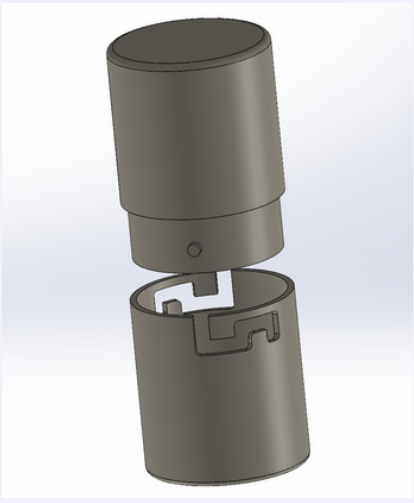
How?

- Designed on **SolidWorks** and tested designs through flight simulation feedback using **OpenRocket**
- Produced plywood rocket parts using wood laser cutter
- Connected parts using two-part epoxy and nails

Results

- Rocket successfully launched, reaching **1054ft**
- Max velocity of **110 m/s**
- Max acceleration of **14.5 G**
- **Successful parachute system deployment** and recovery with **minimal damages**

PAYLOAD HOUSING COMPARTMENT - AERONAUTICS AND ROCKET CLUB







What?

- Protect the payload (large egg) from sustaining any damages during the launch and recovery of the rocket
- Complete creative freedom as long as rocket flight stability is not compromised

How?

- Used **SolidWorks** to design the outer shell, used a bayonet mount system to connect the two halves
- **3D Printing** to produce the outer shell
- Created a melamine foam nest for the egg within the shell
- Surrounded the foam nest with cellulose insulation for heat resistance

Results

- Final compartment fit perfectly inside of rocket body tube
- Successfully **prevented any damage to the egg** during both launch and recovery
- Won **1st place** in the competition for both **best rocket** and **best payload compartment design**